

Check with inverse operations

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Check $7,056 \div 24 = 294$. Copy or complete the first step.

2. Check $4,398 + 6,725 = 11,123$.

3. Check $8,402 - 3,765 = 4,637$.

4. Miro says: Miro repeats the same operation and calls it an inverse check. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Check $7,056 \div 24 = 294$. Copy or complete the first step.
Answer $294 \times 24 = 7,056$, so the quotient is correct.
2. Check $4,398 + 6,725 = 11,123$.
Answer $11,123 - 6,725 = 4,398$.
3. Check $8,402 - 3,765 = 4,637$.
Answer $4,637 + 3,765 = 8,402$.
4. Miro says: Miro repeats the same operation and calls it an inverse check. Is this correct? Explain.
Answer Use the opposite operation so the result returns to the starting value.

Check with inverse operations

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Check $7,056 \div 24 = 294$.

6. Check this result using an inverse, estimate or known fact:
Check $8,402 - 3,765 = 4,637$.

2. Check $4,398 + 6,725 = 11,123$.

7. A division has quotient 318 remainder 7 and divisor 29. Find the dividend.

3. Check $8,402 - 3,765 = 4,637$.

8. Identify and correct this misconception: Miro repeats the same operation and calls it an inverse check.

4. Solve this independently and show every stage: Check $7,056 \div 24 = 294$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Check $4,398 + 6,725 = 11,123$.

10. Write and solve a similar question to: Check $8,402 - 3,765 = 4,637$.

Teacher answers and checking prompts

1. Check $7,056 \div 24 = 294$.
Answer $294 \times 24 = 7,056$, so the quotient is correct.
2. Check $4,398 + 6,725 = 11,123$.
Answer $11,123 - 6,725 = 4,398$.
3. Check $8,402 - 3,765 = 4,637$.
Answer $4,637 + 3,765 = 8,402$.
4. Solve this independently and show every stage: Check $7,056 \div 24 = 294$.
Answer $294 \times 24 = 7,056$, so the quotient is correct.
5. Use a second method or representation for: Check $4,398 + 6,725 = 11,123$.
Answer Expected result: $11,123 - 6,725 = 4,398$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Check $8,402 - 3,765 = 4,637$.
Answer Expected result: $4,637 + 3,765 = 8,402$. A valid check must be shown.
7. A division has quotient 318 remainder 7 and divisor 29. Find the dividend.
Answer 9,229 because $318 \times 29 + 7 = 9,229$.
8. Identify and correct this misconception: Miro repeats the same operation and calls it an inverse check.
Answer Use the opposite operation so the result returns to the starting value.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Check $8,402 - 3,765 = 4,637$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Check with inverse operations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. A division has quotient 318 remainder 7 and divisor 29. Find the dividend.

6. Create a new example that tests the same idea as: Check $8,402 - 3,765 = 4,637$.

2. Prove the answer to this question, rather than only calculating it: Check $7,056 \div 24 = 294$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Check $4,398 + 6,725 = 11,123$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $4,637 + 3,765 = 8,402$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro repeats the same operation and calls it an inverse check.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. A division has quotient 318 remainder 7 and divisor 29. Find the dividend.
Answer 9,229 because $318 \times 29 + 7 = 9,229$.
2. Prove the answer to this question, rather than only calculating it: Check $7,056 \div 24 = 294$.
Answer Expected result: $294 \times 24 = 7,056$, so the quotient is correct. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Check $4,398 + 6,725 = 11,123$.
Answer Expected result: $11,123 - 6,725 = 4,398$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $4,637 + 3,765 = 8,402$.
Answer One valid reconstruction is: Check $8,402 - 3,765 = 4,637$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro repeats the same operation and calls it an inverse check.
Answer Use the opposite operation so the result returns to the starting value.
6. Create a new example that tests the same idea as: Check $8,402 - 3,765 = 4,637$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.